ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CNTT & TT

BÁO CÁO BÀI TẬP
HOME WORK 1

Họ tên: Trịnh Ngọc Quý

Mã số sinh viên: 20235819

Mã lớp: 166155

Hà Nội, Ngày 4 tháng 3 năm 2026

Mục lục

1 Giới thiệu **2**

2 Thiết lập GitHub Repository **2**

3 Bài tập code **2**

3.1 HelloWorld 2

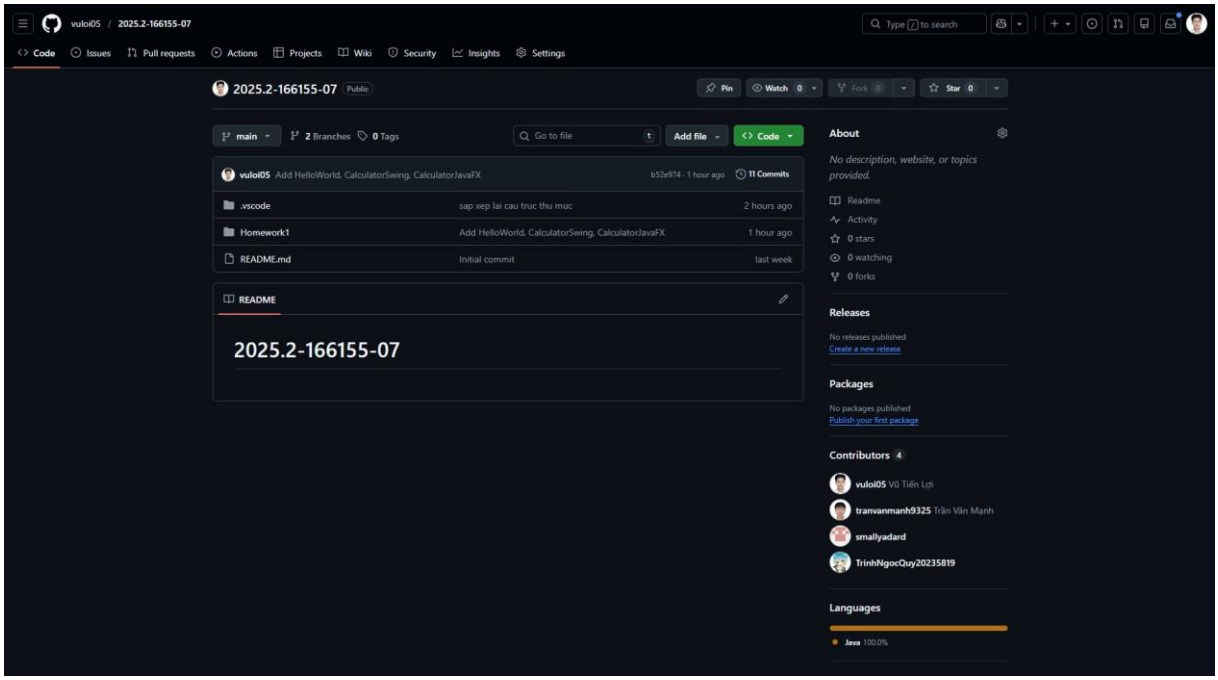
3.2 Calculator Swing (MVC) 3

3.3 Calculator JavaFX (MVC + Scene Builder) 4

4 Thao tác Git **5**

1 Thiết lập GitHub Repository

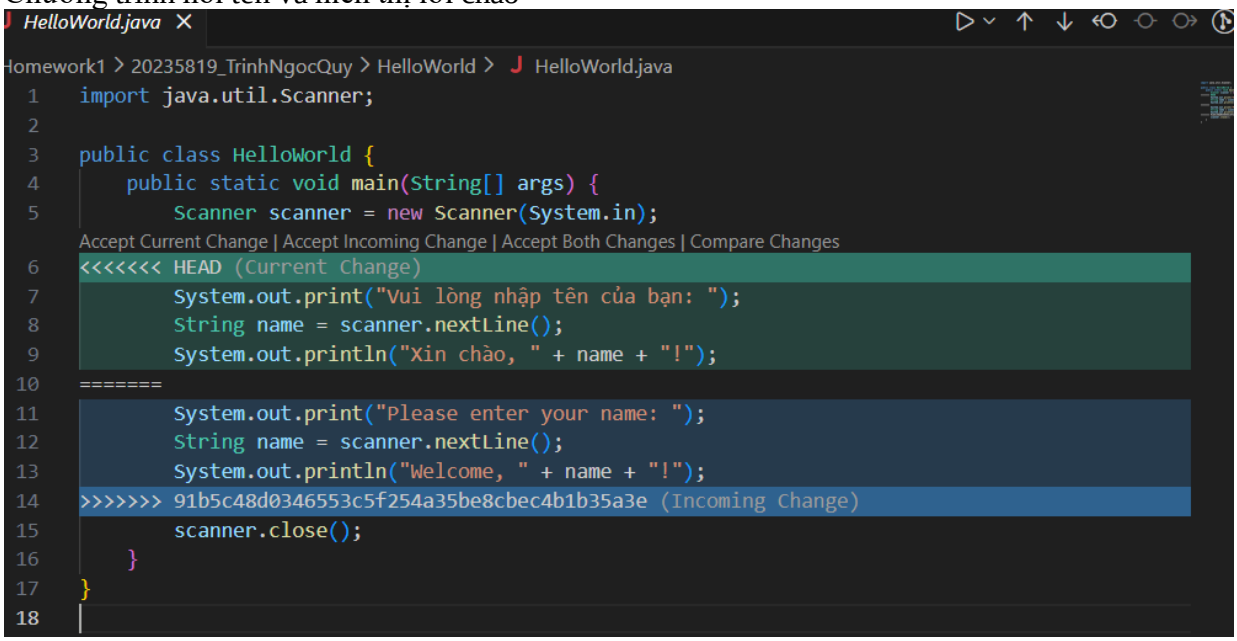
Link đến repository: github.com/vuloi05/2025.2-166155-07



2 Bài tập code

2.1 HelloWorld

Chương trình hỏi tên và hiển thị lời chào



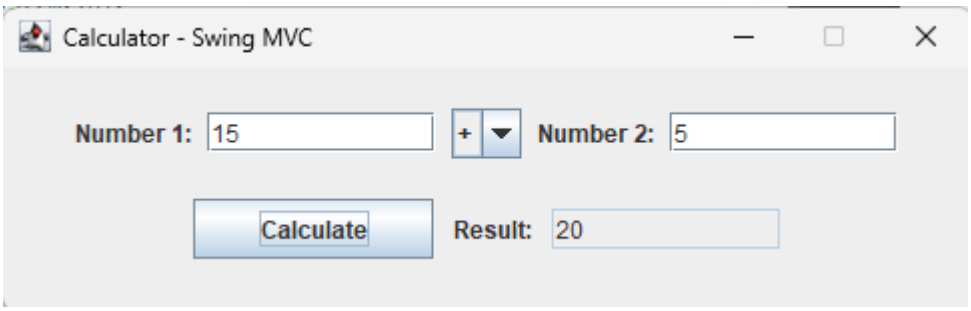
HelloWorld.java

Homework1 > 20235819_TrinhNgocQuy > HelloWorld > HelloWorld.java

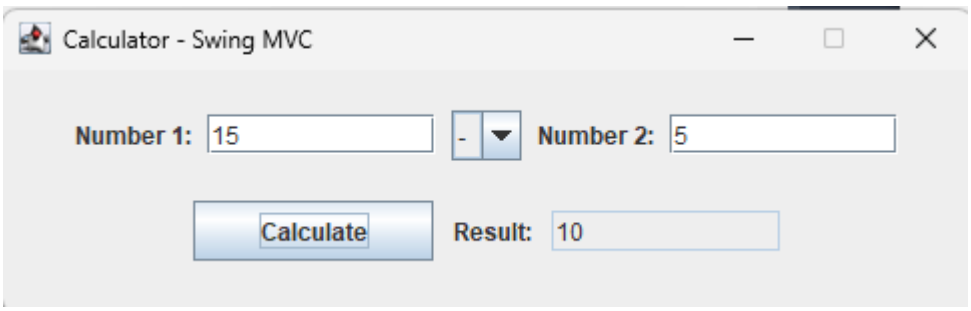
```
import java.util.Scanner;

public class HelloWorld {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<<< HEAD (Current Change)
        System.out.print("Vui lòng nhập tên của bạn: ");
        String name = scanner.nextLine();
        System.out.println("Xin chào, " + name + "!");
=======
        System.out.print("Please enter your name: ");
        String name = scanner.nextLine();
        System.out.println("Welcome, " + name + "!");
>>>>>>> 91b5c48d0346553c5f254a35be8cbec4b1b35a3e (Incoming Change)
        scanner.close();
    }
}
```

2.2 Calculator Swing (MVC)

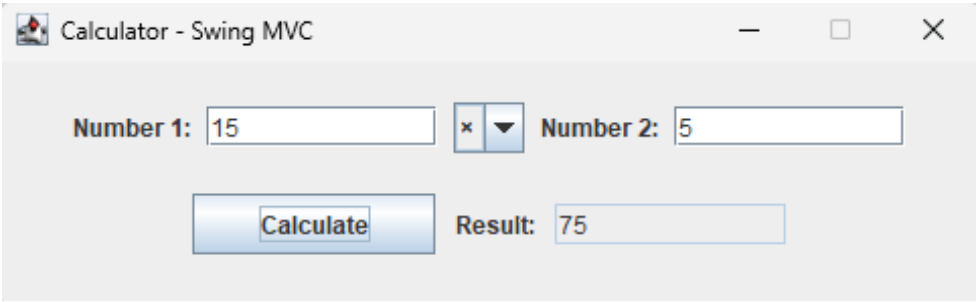
Máy tính 4 phép tính sử dụng Swing + MVC



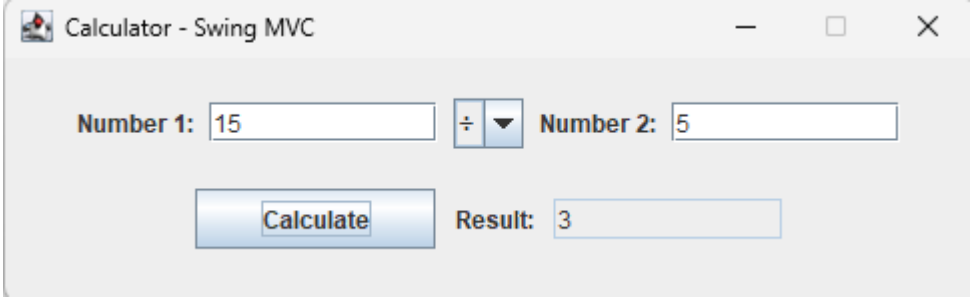
Hình 1: Phép cộng



Hình 2: Phép trừ



Hình 3: Phép nhân



Hình 4: Phép chia

2.3 Calculator JavaFX (MVC + Scene Builder)

Máy tính tương tự nhưng dùng JavaFX và FXML

3 Thao tác Git

```
Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (main)
$ git status
        new file:   HelloWorld/HelloWorld.java

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        BaoCao_HW01.md
        CalculatorJavaFX/
        CalculatorSwing/
        Git_Instructions.md
        MVCTutorial/


Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (main)
$ git commit -m "Thêm chức năng HelloWorld"
[main 250714b] Thêm chức năng HelloWorld
 1 file changed, 11 insertions(+)
 create mode 100644 "HW01_Tr\341\273\213nh Ng\341\273\215c Qu\303\275_20235819/HelloWorld/He
lloWorld.java"

Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (main)
$ git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 665 bytes | 665.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/vuloi05/2025.2-166155-07.git
   074f9f3..250714b  main -> main

Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (main)
$ git branch update-helloworld
Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (main)
$ git branch update-helloworld

Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (main)
$ git checkout update-helloworld
Switched to branch 'update-helloworld'

Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (update-helloworld)
```

```
Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (update-helloworld)
$ git add HelloWorld/HelloWorld.java

Lenovo@LAPTOP-6AJUK6U6 MINGW64 /d/Kỹ năng ITSS/HW01/2025.2-166155-07/HW01_Trịnh Ngọc Quý_202
35819 (update-helloworld)
$ git commit -m "Update loi chao HelloWorld"
On branch update-helloworld
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        BaoCao_HW01.md
        CalculatorJavaFX/
        CalculatorSwing/
        Git_Instructions.md
        MVCTutorial/
```

```
PS C:\DU_AN\2025.2-166155-07> git pull
remote: Enumerating objects: 66, done.
remote: Counting objects: 100% (66/66), done.
remote: Compressing objects: 100% (45/45), done.
remote: Total 64 (delta 5), reused 64 (delta 5), pack-reused 0 (from 0)
Unpacking objects: 100% (64/64), 15.44 KiB | 62.00 KiB/s, done.
From https://github.com/vuloi05/2025.2-166155-07
   63c86ac..9b4d88b  main       -> origin/main
Merge made by the 'ort' strategy.
 Homework 01_Tran_Van_Manh/.idea/workspace.xml      |  87 ++++++++++++++
 Homework 01_Tran_Van_Manh/HelloWorld.java          |  11 ++
 .../calculator_JavaFX/README.md                    | 125 +++++++++++++++++++++++
 .../controller/CalculatorController.java           |  56 +++++++++
 .../calculator_JavaFX/main.java                    |  24 ++++
 .../calculator_JavaFX/model/CalculatorModel.java   |  19 ++++
 .../calculator_JavaFX/pom.xml                      |  81 +++++++++++++
 .../main/java/controller/CalculatorController.java |  56 +++++++++
 .../calculator_JavaFX/src/main/java/main.java      |  24 ++++
 .../src/main/java/model/CalculatorModel.java       |  19 ++++
 .../src/main/resources/view/calculator.fxml        |  31 +++++
 .../classes/controller/CalculatorController.class  | Bin 0 -> 2127 bytes
 .../calculator_JavaFX/target/classes/main.class    | Bin 0 -> 1304 bytes
 .../target/classes/model/CalculatorModel.class     | Bin 0 -> 646 bytes
 .../target/classes/view/calculator.fxml            |  31 +++++
 .../compile/default-compile/createdFiles.lst       |   3 +
 .../compile/default-compile/inputFiles.lst         |   3 +
 .../calculator_JavaFX/view/calculator.fxml         |  32 ++++++
 .../calculator_swing/.vscode/launch.json           |  14 +++
 .../calculator_swing/.vscode/tasks.json            |  40 +++++++
 .../controller/CalculatorController.class          | Bin 0 -> 2085 bytes
 .../controller/CalculatorController.java           |  37 ++++++
 .../calculator_swing/main.class                    | Bin 0 -> 431 bytes
 .../calculator_swing/main.java                     |   8 ++
 .../calculator_swing/model/CalculatorModel.class   | Bin 0 -> 553 bytes
 .../calculator_swing/model/CalculatorModel.java    |  11 ++
 .../calculator_swing/view/CalculatorView.class     | Bin 0 -> 1252 bytes
 .../calculator_swing/view/CalculatorView.java      |  37 ++++++
 .../mvc/controller/HelloController.java            |  20 ++++
 Homework 01_Tran_Van_Manh/mvc/main.java            |  12 ++
 .../mvc/model/HelloModel.java                      |  13 +++
 Homework 01_Tran_Van_Manh/mvc/view/HelloView.java  |  16 +++
 32 files changed, 810 insertions(+)
```

Hình 7: Enter Caption

```
PS C:\DU_AN\2025.2-166155-07> echo "temp" > Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add temp file"
[main 8f25291] Add temp file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git rm Homework1/20235768_VuTienLoi/temp.txt
rm 'Homework1/20235768_VuTienLoi/temp.txt'
PS C:\DU_AN\2025.2-166155-07> git commit -m "Remove temp file"
[main 0bbbc97] Remove temp file
 1 file changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git push origin main
Enumerating objects: 41, done.
Counting objects: 100% (41/41), done.
Delta compression using up to 16 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (35/35), 2.76 KiB | 565.00 KiB/s, done.
Total 35 (delta 16), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (16/16), completed with 3 local objects.
To https://github.com/vuloi05/2025.2-166155-07.git
   56565af..0bbbc97  main -> main
```

Hình 10: Remove file

```
PS C:\DU_AN\2025.2-166155-07> git branch loi-update
PS C:\DU_AN\2025.2-166155-07> git checkout loi-update
Switched to branch 'loi-update'
PS C:\DU_AN\2025.2-166155-07> git branch
* loi-update
  main
PS C:\DU_AN\2025.2-166155-07>
```

Hình 11: Create branch

```
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Update greeting message on feature branch"
[loi-update c5f05e8] Update greeting message on feature branch
 1 file changed, 1 insertion(+), 2 deletions(-)
PS C:\DU_AN\2025.2-166155-07> git push origin loi-update
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 639 bytes | 639.00 KiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'loi-update' on GitHub by visiting:
remote:      https://github.com/vuloi05/2025.2-166155-07/pull/new/loi-update
remote:
To https://github.com/vuloi05/2025.2-166155-07.git
 * [new branch]      loi-update -> loi-update
```

Hình 12: Thay đổi file

```
PS C:\DU_AN\2025.2-166155-07> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\DU_AN\2025.2-166155-07> git merge feature-update
merge: feature-update - not something we can merge
PS C:\DU_AN\2025.2-166155-07>
```

Hình 13: Gộp nhánh

```
HelloWorld > src > HelloWorld.java > ...
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<<< HEAD    Syntax error on tokens, delete these tokens (Current Change)
// Main branch version
=======
// Conflict branch version
>>>>>>> conflict-branch-loi (Incoming Change)

import java.util.Scanner;

public class HelloWorld {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print(s: "Please enter your name: ");
        String name = scanner.nextLine();

        System.out.println("Hello, " + name + "!");

        System.out.println(x: "Welcome to Homework01!");

        scanner.close();
    }
}
```

Hình 14: Gộp nhánh có lỗi - 1

```
HelloWorld > src > HelloWorld.java > ...
// Main branch version    HelloWorld.java is a non-project file, only syntax errors are reported

import java.util.Scanner;

public class HelloWorld {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print(s: "Please enter your name: ");
        String name = scanner.nextLine();

        System.out.println("Hello, " + name + "!");

        System.out.println(x: "Welcome to Homework01!");

        scanner.close();
    }
}
```

Hình 15: Gộp nhánh có lỗi - 2

```
PS C:\DU_AN\2025.2-166155-07> git branch
  conflict-branch-loi
  loi-update
* main
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add comment on main branch"
[main 787e387] Add comment on main branch
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\DU_AN\2025.2-166155-07> git checkout -b conflict-branch-loi
fatal: a branch named 'conflict-branch-loi' already exists
PS C:\DU_AN\2025.2-166155-07> git checkout conflict-branch-loi
Switched to branch 'conflict-branch-loi'
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add comment on conflict branch"
On branch conflict-branch-loi
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Homework1/20235768_VuTienLoi/.vscode/
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/Main.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/controller/CalculatorController$CalculateListener.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/controller/CalculatorController.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/model/CalculatorModel.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/view/CalculatorView.class

nothing added to commit but untracked files present (use "git add" to track)
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add comment on conflict branch"
[conflict-branch-loi 3174364] Add comment on conflict branch
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\DU_AN\2025.2-166155-07> git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)
PS C:\DU_AN\2025.2-166155-07> git merge conflict-branch
merge: conflict-branch - not something we can merge
PS C:\DU_AN\2025.2-166155-07> git merge conflict-branch-loi
Auto-merging Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
CONFLICT (content): Merge conflict in Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
Automatic merge failed; fix conflicts and then commit the result.
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Resolve merge conflict"
[main b72bb62] Resolve merge conflict
PS C:\DU_AN\2025.2-166155-07>
```

Hình 16: Gộp nhánh có lỗi - 3